



DEALER BEST PRACTICE SERIES

3500 Engine Cylinder Head Bolt Torque Fixture

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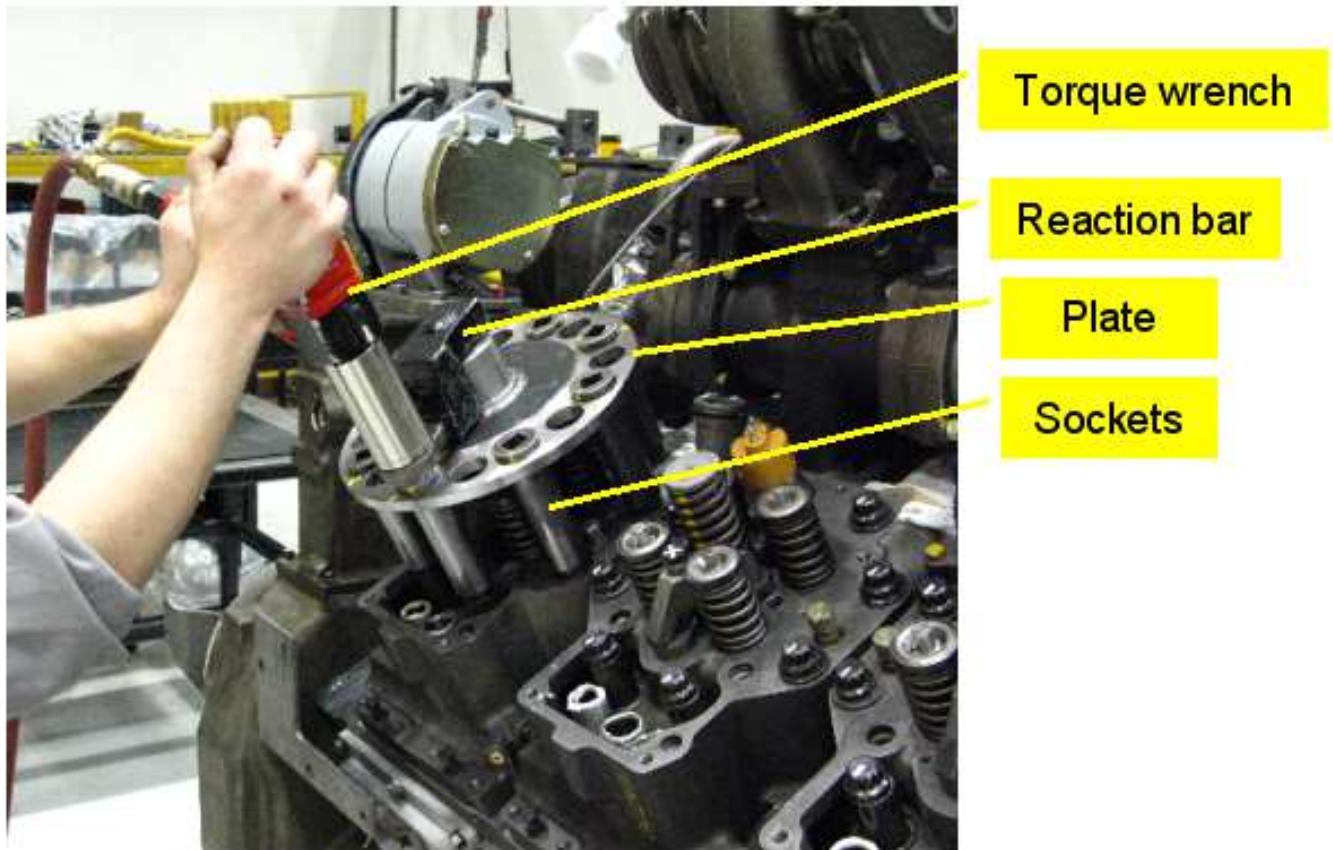
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1.0 Introduction

This Best Practice is a dealer-fabricated fixture to improve the process to set 3500 engine cylinder head bolt torque. The fixture allows the use of a pneumatic torque wrench with proper alignment and engagement of the sockets on the head bolts and support of the torque wrench and reaction arm.

The fixture reduces the time required to torque all the cylinder head bolts by 20% or more, reduces technician fatigue, and improves the safety of the procedure.



2.0 Best Practice Description

The purpose of this fixture is to improve the process for setting cylinder head bolt torques on 3500 series engines.

The fixture is mounted on the cylinder head with all eight sockets engaged on the head bolts. A pneumatic torque wrench is used to torque the bolts individually following the Service Manual sequence and initial torque. The reaction arm of the torque wrench is a pin that is secured in the center of the fixture. This allows easy access to all the bolts.

After initial torque, the bolt head position is marked on the fixture and the final “torque turn” is performed.

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Note: The head bolts are torqued before the fuel injector is installed in the cylinder head. Fixture does not have spring clearance with the injector installed.



After the initial torque has been set, the operator marks the plate with a felt pen and completes the final torque turn.

3.0 Implementation Steps

The first step is to fabricate the steel top plate of the fixture. Using a Caterpillar print, holes are drilled in the plate matching the cylinder head bolt hole pattern. See drawing below.

Sockets are slip fitted into the plate hole pattern and retained by a flange on the top of the socket and a snap ring on the under side of the plate.

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Socket diameters will need to be machined to create a flange at the top of the socket. The groove is also machined in the sockets to hold a snap ring. The sockets are slip fitted into the plate and held in position by the flange and the snap ring.

Sockets used are Cat part number 124-0662. This is a $\frac{3}{4}$ " 12PT socket – 5" long. The diameter of the socket is machined to provide .003-.005" clearance through the plate.



Sockets have been machined leaving a shoulder on this side of the plate.

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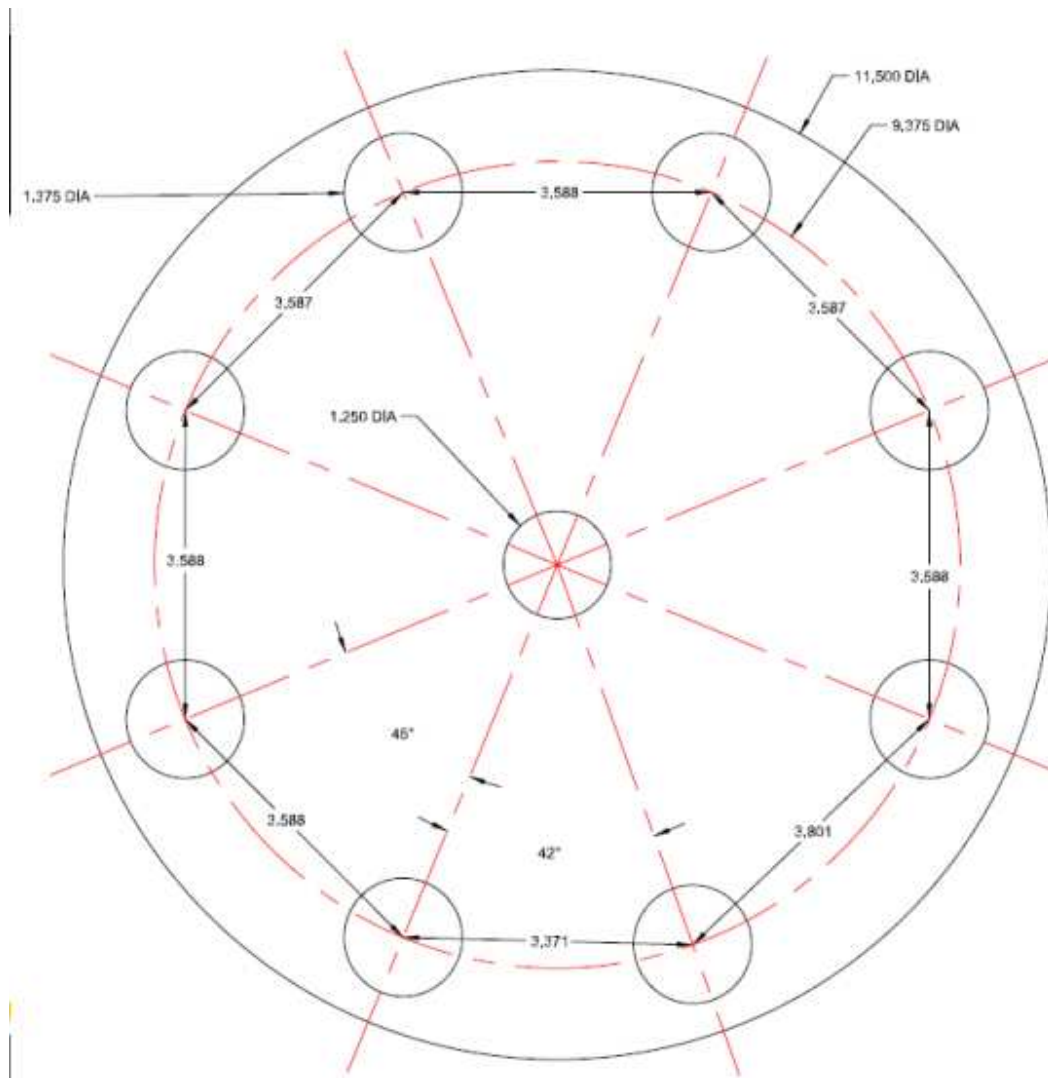


A torque wrench and a reaction arm for the torque wrench will need to be manufactured. It is best to work with your torque wrench supplier to match a reaction arm and wrench to this fixture.



Torque wrench is a Norbar PTM52 Model 18101B06. Range is 147-735 ft/lbs, $\frac{3}{4}$ " drive.

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It may be beneficial to make several fixtures as well as spare parts (sockets, reaction arm). Keep your drawings and specifications for potential future replication of the tool if needed.

4.0 Benefits

- Reduces time required to torque head bolts 20% or more.
- Less fatigue on technician.
- Better torque accuracy.
- Less chance of socket coming off the bolt during torque – improve technician safety.

5.0 Resources Required

- Machine shop and time to fabricate. Approximate cost of the fixture is US \$1500.
- Cost of pneumatic wrench, air regulator, and reaction arm approximately US\$7500.

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6.0 Supporting Attachments / References

None

7.0 Related Best Practices

None at this time.

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